# Introduction to Generative AI Applications

## Practical Generative AI Use Cases

### Introduction to Generative AI:

Generative AI refers to a category of artificial intelligence systems that can create new content  such as audio, video, text or other data rather than just analysing and classifying existing data. Generative AI models learn structure and patterns of large amount of data and use that understanding to produce new human like output. The most common type of generative AI is LLM (large language model) which is GPT, claude ai, gemini etc. These models are trained on vast amounts of data to understand patterns, styles, and facts. When a prompt is given, an LLM generates a relevant response by predicting the most likely sequence of words.

## Use Case 1: AI Email Assistant

**Problem Statement:** Government office staff spend alot of time by writing the same type of email repeatedly, like replying to citizens' complaints or asking for documents. This takes time and slows down the work.

**Target User:** Administrative staff across various government departments.

**Input:** The email received from a citizen along with the key information that should include in reply like processing takes 10 days or we have received your request.

**AI Process:** The AI reads citizens email and all necessary information and then writes a professional email in proper format.

**Expected Output:** A ready-to-send professional email with correct tone and no missing facts

**Limitations:** AI might not know the new policies unless they are provided. It may reply too general to legal or official matters so one person must check it before sending it.

**Possible Improvement:** Connect the AI with the department's latest policy documents so it can always give correct and up-to-date information.

**Prompt:** Write a formal reply email from the Ministry of Education to a student who is asking why their scholarship has not yet been granted, even though they applied a few months ago. Explain that the process takes time because the submitted documents must first be verified, and then the Ministry needs to receive confirmation from the student's university before the scholarship can be approved. Apologize for the delay, reassure the student that their application is being processed, and ask them to wait patiently. Keep the email under 150 words. Use a professional tone with a proper greeting, closing, and official email format.

## Use Case 2: AI Report Summarizer

**Problem Statement**: Government departments receives long technical or statistical reports but rarely have time to read them in full before meetings, risking uninformed decisions

**Target User:** Senior officials, department heads in public sector office

**Input:** A long report such as a quarterly budget report.

**AI Process:** The AI reads the report and produces an executive summary with proper headings and sections and page numbers.

**Expected Output:** A one-page summary with proper headings, sections and page numbers.

**Limitations:** If the AI leaves out important risk or point in the summary, it may not be clear who is responsible human reviewer or the AI, which creates an accountability issue. AI may not understand tables or chart, it may misinterpret them, affecting accuracy.

**Possible Improvement:** Add note to the summary that clearly stating what was not included and have a person to review it before sharing it. If AI is uncertain about something it should flag it instead of guessing it

**Prompt:** Summarize this quarterly budget report. Organize the summary with headings for each department. Include the page number for every key statistic. Keep the summary under 400 words and end it with a short conclusion.

## Use Case 3: AI Meeting Notes Generator

**Problem Statement:** Public meetings, whether held in person or online (such as internal briefing) often go undocumented, making it hard to track decisions and follow ups later.

**Target User:** Secretaries and official administrators

**Input:** Input can be rough meeting notes, a written transcript, or a transcript generated from a recorded online meeting.

**AI Process:** The AI organizes these meeting transcripts into structured minutes, with topic discussed and decisions made. If recording is provided audio first converted to text using speech to text transcription, the resulting transcription is then organized by llm into structured minutes.

**Expected Output:** A formatted minutes document with clear sections and a list of action items.

**Limitations:** Some meetings include private information such as staff and budget discussions. Sending these meeting transcripts to an ai tool may create privacy and confidentiality risks. Video transcript may introduce errors from background noises when it converted to text .

**Possible Improvement:** Only use AI for meetings that do not contain sensitive information and do not use any tool that stores or reuse data. We can connect AI to the online meeting platforms like zoom so it can automatically generate transcript.

**Prompt:** Read this meeting transcript and convert it into formal meeting minutes. Include the meeting title (if available), date (if mentioned), attendees, topics discussed, key decisions, action items with the responsible person and deadline (if mentioned), and a summary of the important points discussed by each attendee. Use professional English, organize the information under clear headings, and present the important points in bullet form. Do not add any information that is not mentioned in the transcript.

## Use Case 4: AI Document Q&A Assistant

**Problem Statement:** Government staff and citizens often need quick answers from lengthy policy documents, but searching manually is slow and error prone.

**Target User:** Citizens and government staff

**Input:** A policy document and a specific question.

**AI Process:** The AI searches the document and answers the question using only the information available in the document.

**Expected Output:** A direct, accurate answer with a reference to where in the document it was found.

**Limitations:** It may guess an answer if it is not present in the document and very large documents may be difficult to process.

**Possible Improvement:** AI models tell clearly that this part is not available in this document instead of guessing.

**Prompt:** Read this attach document and answer this question: What did the total utilization rate this quarter, and what are the department's recommendations? List the recommendations in bullet points.

## Use Case 5: AI Idea Generator for Public Services

**Problem Statement:** Government offices often need new ideas to improve public services but may not have a proper brainstorming process.

**Target User:** Policy planners / innovation teams in government departments.

**Input:** Description of problem that need to be resolved like long queues at passport office.

**AI Process:** The AI generates multiple improved ideas to solve problem and explain how practical each idea is.

**Expected Output:** A list of 5–7 useful ideas with simple notes about cost and implementation.

**Limitations:** Ideas may be too generic. The AI does not know actual budget it might suggest ideas that don't fit financially or it might also suggest ideas that are already tried.

**Possible Improvement:** Feed the AI historical data on past initiatives so it avoids repeating solutions that already failed.

**Prompt:** Suggest 5 simple and practical ideas to improve waste collection and cleanliness in a city. For each idea, write one short explanation and mention whether it is low, medium, or high cost and whether it is easy or difficult to implement.